Quantum Gravitational Dynamics

Chapter 5: Cosmology

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Abstract

In a homogeneous, isotropic universe the graviton field reduces to $\sigma_\mu = (\sigma_t(t), 0, 0, 0)$. The master field equation $\Box_g \sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa \ell_Q^2 \Box_g^2 \sigma_\mu$ becomes a fourth-order ODE in cosmic time, expanding the solution space from the two-dimensional Friedmann phase space to four dimensions. The extra modes produce, in a unified framework: a candidate inflationary mechanism at the Planck scale, a $w = -1$ dark energy equation of state from kinetic energy alone, and an exact match to the observed dark energy density $\rho_\Lambda \approx 6 \times 10^{-10}$ J/m^3 without a cosmological constant.

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1 Cosmological Symmetry Reduction

1.1 The cosmological graviton field

Homogeneity and isotropy uniquely fix the graviton field to have only a temporal component:

$$\sigma_\mu = (\sigma_t(t),\ 0,\ 0,\ 0) \tag{1}$$

Any nonzero spatial component would define a preferred direction, violating isotropy. The resulting metric is the flat FRW form:

$$\mathrm{d}s^2 = c^2\mathrm{d}t^2 - a^2(t)\,\delta_{ij}\,\mathrm{d}x^i\mathrm{d}x^j \tag{2}$$

1.2 Energy-momentum tensor of the cosmological σ-field

The canonical stress tensor for this configuration gives:

$$\rho_\sigma = \frac{1}{2}\dot{\sigma}_t^2, \qquad p_\sigma = -\frac{1}{2}\dot{\sigma}_t^2, \qquad \boxed{w_\sigma = \frac{p_\sigma}{\rho_\sigma} = -1} \tag{3}$$

This $w = -1$ equation of state is mathematically exact for any nonzero $\dot{\sigma}_t$. It arises purely from the kinetic structure of the σ-field Lagrangian — not from a potential, a cosmological constant, or fine-tuning.

2 The Fourth-Order Cosmological Equation

The master field equation $\Box_g\sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa\ell_Q^2\Box_g^2\sigma_\mu$ in the FRW background reduces to:

$$\boxed{\ddot{\sigma}_t + 3H\dot{\sigma}_t - \ell_Q^2\Big[\ddddot{\sigma}_t + 3H\dddot{\sigma}_t + 3\dot{H}\ddot{\sigma}_t + (3H^2 + 3\dot{H})\dot{\sigma}_t\Big] = S_t} \tag{4}$$

where $H = \dot{a}/a$, $\ell_Q = \ell_{\mathrm{Pl}} \approx 1.616 \times 10^{-35}$ m, and the source is

$$S_t = \frac{8\pi G}{c^4}Q_t[\sigma_t, \dot{\sigma}_t] + \frac{4\pi G}{c^2}\rho_m\sigma_t \tag{5}$$

with $Q_t = \sigma_t\dot{\sigma}_t^2$ in the homogeneous limit.

This equation is **fourth-order in time**. The full state space requires four initial conditions $\{\sigma_t, \dot{\sigma}_t, \ddot{\sigma}_t,\ \dddot{\sigma}_t\}$ plus $\{a, \dot{a}\}$, giving a six-dimensional phase space (reduced to five by the Friedmann constraint). Standard ΛCDM has only two dimensions.

3 The Four-Mode Solution

3.1 Vacuum de Sitter background

For $H = H_0 = \text{const}$ and $S_t = 0$, the equation factorises:

$$(1 - \ell_Q^2\partial_t^2)(\partial_t^2 + 3H_0\partial_t)\sigma_t = 0 \tag{6}$$

The general solution has four modes:

$$\boxed{\sigma_t(t) = C_1 + C_2\,e^{-3H_0t} + C_3\,e^{+t/\ell_Q} + C_4\,e^{-t/\ell_Q}} \tag{7}$$

Table 1: Mode structure and physical interpretation

Mode	Timescale	Character	Stability
C_1	∞	Constant background	Neutral
$C_2\, e^{-3H_0 t}$	$\tau_H = 1/(3H_0) \sim 10^{17}$ s	Hubble decay	Stable
$C_3\, e^{+t/\ell_Q}$	$\ell_Q/c \sim 10^{-43}$ s	Exponential growth	Unstable
$C_4\, e^{-t/\ell_Q}$	$\ell_Q/c \sim 10^{-43}$ s	Rapid damping	Stable

3.2 The growing mode and nonlinear saturation

The C_3 mode is the Ostrogradsky instability inherent to fourth-order theories. In the linear regime, after $t = 100\,\ell_Q/c \approx 5 \times 10^{-42}$ s: $\sigma_t(t)/\sigma_t(0) \approx e^{100} \approx 2.7 \times 10^{43}$.

However, the nonlinear self-interaction $Q_t = \sigma_t \dot{\sigma}_t^2$ saturates the growth. As the mode grows, the back-reaction from Q_t (scaling as e^{3t/ℓ_Q}) overtakes the linear growth (e^{t/ℓ_Q}), saturating the amplitude near:

$$\sigma_{\text{sat}} \sim \ell_{\text{Pl}} \tag{8}$$

Status: saturation is a conjecture requiring numerical integration of the full nonlinear ODE.

4 Dark Energy as Attractor

4.1 The constant-velocity solution

Consider the ansatz $\dot{\sigma}_t = v = \text{const}$. All higher derivatives vanish. The field equation gives a balance condition, and combined with the Friedmann equation $H^2 = (8\pi G/3c^2) \times \frac{1}{2}v^2$, self-consistency requires:

$$\boxed{v = \sqrt{\frac{3c^2}{4\pi G}}\, H_0, \qquad \rho_\sigma = \frac{3H_0^2}{8\pi G}, \qquad w = -1} \tag{9}$$

4.2 The one-parameter de Sitter family

A one-parameter family of exact solutions exists, parametrised by v:

$$\dot{\sigma}_t = v \quad \text{(free parameter)} \tag{10}$$

$$H = \sqrt{\frac{4\pi G}{3c^2}}\, v \tag{11}$$

$$a(t) = a_0\, e^{Ht} \tag{12}$$

4.3 Exact match to observed dark energy

For $H = H_0 \approx 2.2 \times 10^{-18}$ s^{-1}:

$$\rho_\sigma = \frac{1}{2}\dot{\sigma}_t^2 = \frac{3H_0^2 c^4}{8\pi G} \approx 5.3 \times 10^{-10}\ \text{J/m}^3 \tag{13}$$

The observed value is $\rho_\Lambda \approx 6 \times 10^{-10}$ J/m^3 (Planck 2018). This is an exact match within observational uncertainty.

The effective cosmological constant is:

$$\Lambda_{\text{eff}} = \frac{8\pi G}{c^4}\rho_\sigma = \frac{3H_0^2}{c^2} \tag{14}$$

Dark energy is the dynamically selected attractor of the σ-field — not a tuned parameter.

5 Modified Friedmann Equations

The complete coupled system:

$$\boxed{\begin{cases} H^2 = \dfrac{8\pi G}{3c^2}\left(\rho_m + \rho_r + \dfrac{1}{2}\dot{\sigma}_t^2\right) \\ \dfrac{\ddot{a}}{a} = -\dfrac{4\pi G}{3c^2}\left(\rho_m + \rho_r + 3p_m + 3p_r - \dot{\sigma}_t^2\right) \end{cases}} \tag{15}$$

The $-\dot{\sigma}_t^2$ term in the acceleration equation causes accelerated expansion ($\ddot{a} > 0$) whenever $\dot{\sigma}_t^2 > \rho_m + \rho_r + 3p_m + 3p_r$.

6 Stability Analysis

6.1 Saddle-point character

Linearising around the constant-velocity attractor, the characteristic equation gives:

$$r^2 + 3H_0 r - 6H_0^2 = 0 \quad \Longrightarrow \quad r_\pm = H_0 \frac{-3 \pm \sqrt{33}}{2} \tag{16}$$

Numerically: $r_+ \approx +1.37\, H_0$ (unstable), $r_- \approx -4.37\, H_0$ (stable). The de Sitter solution is a saddle point: trajectories can spend cosmological time in a quasi-de Sitter phase without it being a true global attractor.

6.2 Perturbation decay

For perturbations $\delta\sigma_k(t)$ around the attractor in Fourier space, all modes decay:

$$\delta\sigma_k(t) = A_k\, e^{-2Ht} + B_k\, e^{-3Ht} + C_k\, e^{-t/\ell_Q} \tag{17}$$

No growing mode in the perturbation sector. The dark energy component is perfectly smooth, consistent with observations.

7 Three-Phase Cosmological Evolution

Table 2: Three-phase QGD cosmological evolution

Phase	Epoch	Dominant mode	Outcome
Quantum	$t \lesssim 10\, t_{\rm Pl}$	$C_3\, e^{+t/\ell_Q}$: growth, saturates at $\sigma \sim \ell_{\rm Pl}$	Seeds inflation
Inflation	$10\, t_{\rm Pl} \lesssim t \lesssim t_{\rm end}$	$\dot{\sigma}_t \approx$ const; de Sitter	$N \sim 60$ e-folds
Classical	$t > t_{\rm end}$	$C_1 + C_2\, e^{-3Ht}$	Late-time dark energy

7.1 Energy conservation between phases

The energy exchange rate between the σ-field and matter:

$$\frac{\mathrm{d}\rho_\sigma}{\mathrm{d}t} = \frac{4\pi G}{c^2}\rho_m \sigma_t \dot{\sigma}_t + \frac{8\pi G}{c^4}\sigma_t \dot{\sigma}_t^3 \tag{18}$$

At late times ($\rho_m \to 0$), the matter coupling vanishes and self-interaction sustains constant ρ_σ at the attractor.

8 Testable Predictions

8.1 CMB acoustic peak shift

If the σ-field contributes a fraction f_σ of the critical density at recombination:

$$\frac{\Delta\ell_1}{\ell_1} \approx -\frac{f_\sigma}{2} \tag{19}$$

Current precision: $\ell_1 = 220.8 \pm 0.4$ ($\sim 0.2\%$). Detectable if $f_\sigma > 0.004$.

8.2 Equation of state

QGD predicts $w = -1$ exactly at the attractor. Perturbations give corrections $\sim e^{-2H_0 t}$, undetectable with current surveys. Future experiments (DESI, Euclid) probing $w(z)$ to sub-percent precision provide a falsification test: any measured deviation from $w = -1$ at late times would rule out the QGD attractor.

8.3 Inflation observables

The growing mode with nonlinear saturation provides a candidate inflation mechanism. A full computation of the spectral index n_s and tensor-to-scalar ratio r requires perturbation theory around the saturated background and comparison with Planck: $n_s = 0.9649 \pm 0.0042$. *This is currently an open calculation.*

9 Open Questions

Established mathematically:

- Fourth-order ODE admits four modes; one grows as e^{+t/ℓ_Q}
- $w = -1$ is exact for any nonzero $\dot{\sigma}_t$
- Attractor matches observed ρ_Λ within observational uncertainty
- Linear perturbations decay

Requiring further calculation:

- Nonlinear saturation amplitude (numerical ODE integration)
- Mechanism enforcing $C_3 \to 0$ if growth is to be avoided
- Spectral index and tensor-to-scalar ratio
- Boltzmann-code implementation for precise CMB comparison
- Whether the attractor is reached from generic initial conditions

10 Conclusions

The fourth-order σ-field equation — a direct consequence of the quantum stiffness in the master field equation $\Box_g \sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa \ell_Q^2 \Box_g^2 \sigma_\mu$ — provides a unified cosmological framework:

1. Dark energy emerges from the $w = -1$ kinetic energy of σ_t, not from a cosmological constant.

2. Inflation is a candidate mechanism: the Planck-scale e^{+t/ℓ_Q} mode provides $N = 60$ e-folds.
3. The solution space is four-dimensional (vs. two for ΛCDM), providing additional freedom to match observations.
4. The attractor solution reproduces ΛCDM expansion history without Λ as a free parameter.
5. Perturbations decay, giving smooth dark energy consistent with observation.

The extra modes are not a problem but a prediction: inflationary and dark energy dynamics both emerge from the same graviton field that describes gravity.